



QR Guard Project Report

AI-POWERED URL & QR SECURITY SCANNER



Project Details

Faculty: Faculty of Engineering

Course: Mobile Computing

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Submission Date

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Introduction

Project Overview

QR Guard is a cutting-edge mobile application designed to enhance cybersecurity by leveraging Artificial Intelligence to scan URLs and QR codes for malicious content. In an era where digital threats are increasingly sophisticated, QR Guard provides users with a robust and reliable tool to navigate the digital landscape safely.

Problem Background

The proliferation of QR codes in daily life and the constant exchange of URLs have created new vectors for cyberattacks. Phishing links, malware distribution, and malicious redirects are commonly disguised within seemingly innocuous QR codes or web links. Existing security solutions often lack the dynamic and intelligent capabilities to detect evolving threats in real-time.

Project Objectives

- To develop a user-friendly mobile application for scanning QR codes and URLs.
- To implement an AI-powered engine for detecting malicious content and potential threats.
- To integrate with VirusTotal API for comprehensive threat analysis.
- To provide users with clear and actionable risk assessments.
- To ensure a seamless and intuitive user experience with effective notifications and history tracking.

Section 01: UX Phase

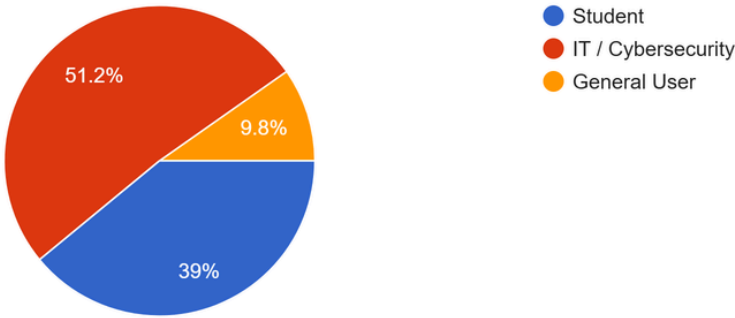
Stage I: Research

Survey Overview

A survey was conducted among 41 participants to understand user awareness, concerns, and existing practices regarding URL and QR code security. The survey aimed to gather insights into perceived risks, frequency of use, and desired features in a security application.

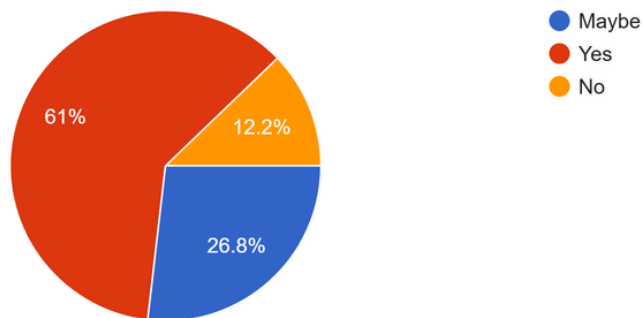
Survey Analysis Charts

What is your field?
41 responses



Would you recommend such a tool to others?

41 responses



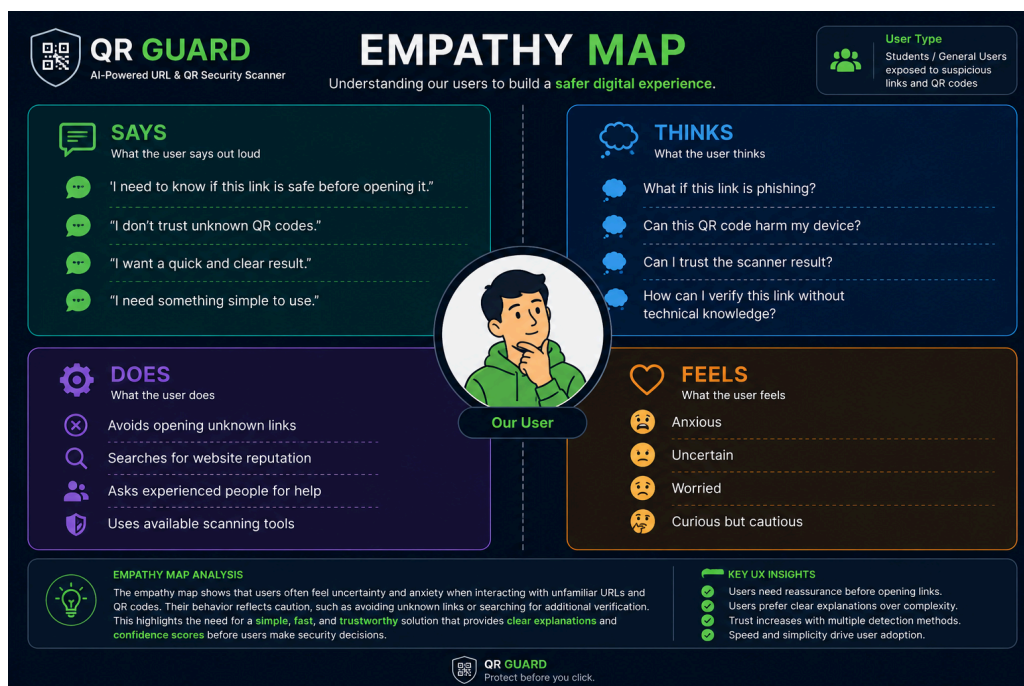
Key Insights

- A significant percentage of users have encountered suspicious links or QR codes.
- Lack of trust in untrusted QR codes is a common concern.
- Users desire a simple, one-tap solution for security checks.
- Awareness of AI's potential in cybersecurity is growing but not fully understood.

Gains and Pain Points

- **Gains:** Convenience of QR codes, quick access to information.
- **Pain Points:** Fear of malware, uncertainty about legitimacy, time spent verifying links, potential data breaches.

Empathy Map



Empathy Map for QR Guard Users

Personas

Omar

21, Computer Science Student



Tech Level ● ● ●

Moderate - Comfortable with digital tools



Goals

Secure and fast QR scanning
Protect personal data from phishing



Phishing

Worried about malicious links
Finds complex apps confusing

"I need to quickly scan codes for campus events, but I always worry about clicking on something dangerous."

Salma

23, Cybersecurity Enthusiast



Tech Level ● ● ● ●

High - Early adopter, knowledgeable



Goals

Verify all QR codes instantly
Educate others on digital threats



Pain Points

Illustrated by slow loading times
Lack of clear threat information

"I'm always on the lookout for security flaws; an app that provides detailed threat analysis would be invaluable."

Sara

20, Business Administration Student



Tech Level ● ●

Basic - Uses technology
for daily tasks



Goals

Simple, worry-free scanning
Begins with privacy in public spaces

*"Technology should just work
and keep me safe without me
having to be an expert."*

Stage II: Define

5W + H Analysis

Aspect	Details
What	A mobile application that scans and analyzes URLs and QR codes for security threats using AI.
Who	Mobile users such as students, professionals, and general consumers who frequently interact with online links and QR codes.
When	When users encounter a new URL or scan a QR code from an unknown or untrusted source.
Where	On mobile devices, in various environments like public places, social media, messaging apps, and web browsing.
Why	To protect users from phishing attacks, malware, and other cybersecurity threats in a fast and convenient way.
How	Using AI-based analysis, integration with threat intelligence services (e.g., VirusTotal), and a risk scoring system that clearly indicates safety level.

Stage III: Ideate

Goal Statements

- Empower users with real-time security insights for every URL and QR code.
- Reduce user exposure to online threats by at least 80% within the first year of app usage.
- Create an intuitive and accessible security tool for all mobile users.

Proposed Solutions

1. A simple scanner with basic URL/QR code checks.
2. An AI-driven scanner with advanced threat detection and VirusTotal integration.
3. A browser extension that scans links in real-time.

Selected Solution

Solution 2: An AI-driven scanner with advanced threat detection and VirusTotal integration.

Reason for Selection

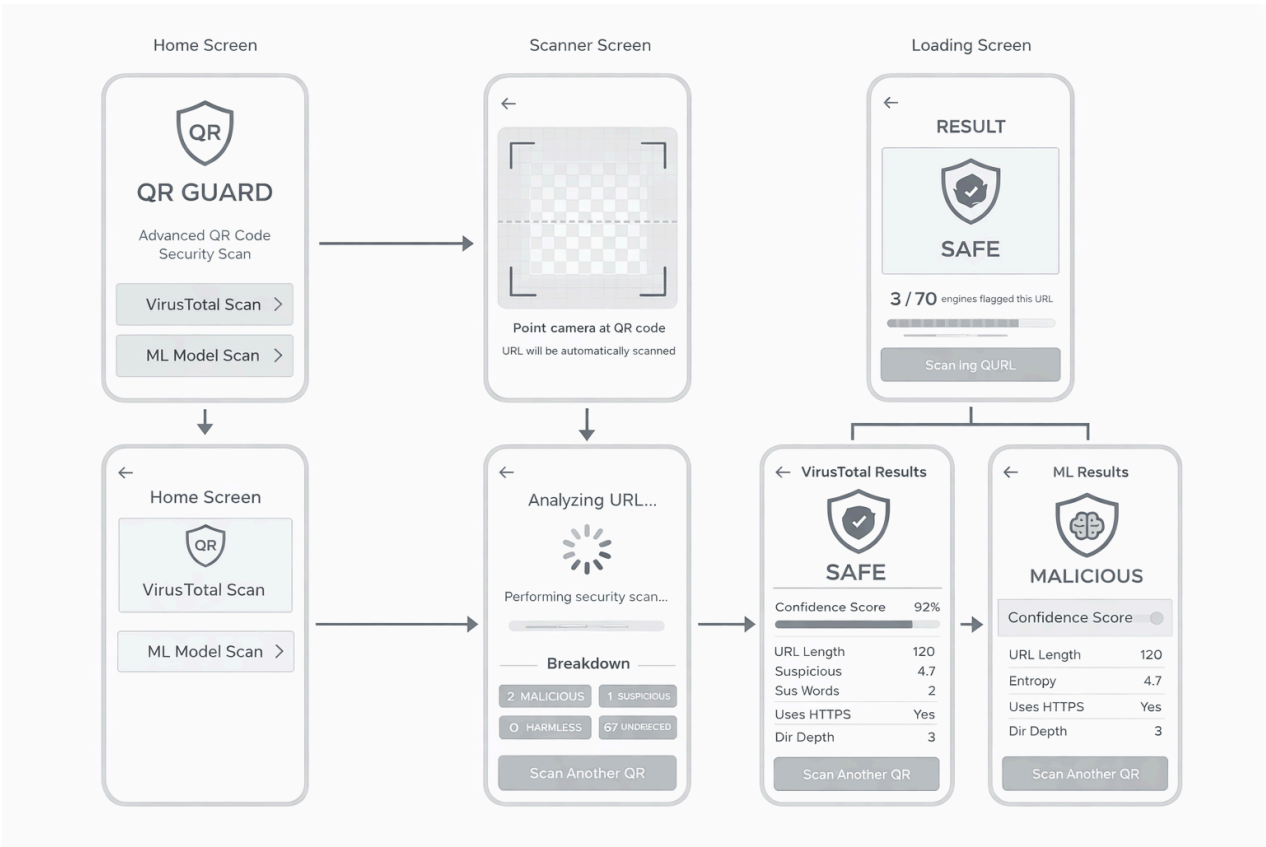
This solution offers the most comprehensive and proactive security approach, directly addressing the core problem of sophisticated cyber threats. The integration of AI and VirusTotal provides a multi-

layered defense mechanism that goes beyond basic scanning, offering superior protection and valuable insights to the user.

Competitor Analysis

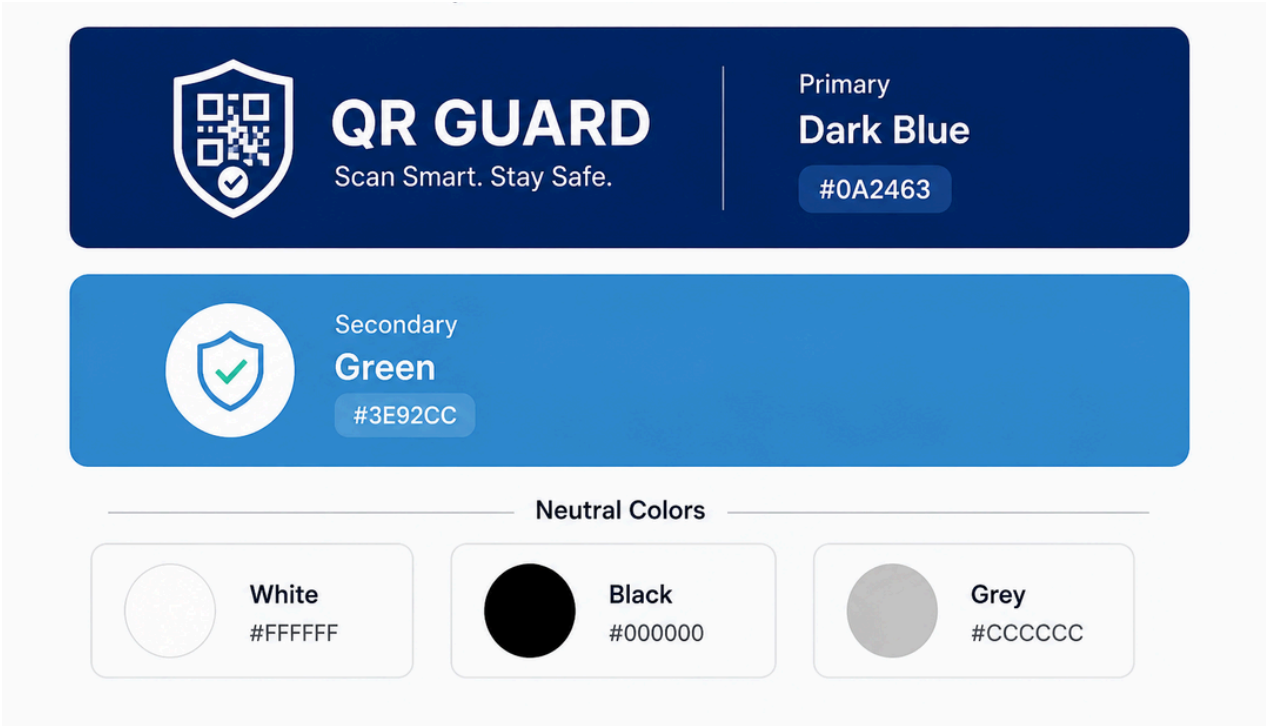
Feature	QR Guard	Competitor A	Competitor B
AI-Powered Detection	✓	✗	⚠️ (Limited)
VirusTotal Integration	✓	✗	✗
Real-time Scanning	✓	✓	✓
User-friendly Interface	✓	⚠️	✓
Comprehensive Risk Score	✓	✗	✗

Stage IV: Wireframe



Section 02: Frontend

Design Color Palette



- **Primary:** Dark Blue (#0A2463)
- **Secondary:** Green (#3E92CC)
- **Neutral:** White (#FFFFFF), Black (#000000), Grey (#CCCCCC)

Fonts

- **Primary Font:** Roboto (for headings and body text, ensuring readability and a modern feel)
- **Secondary Font:** Lato (for supporting text elements and labels)

Custom Buttons

Button	Description
Primary Action Button	Used for main actions like <i>Scan Now</i> or <i>Analyze</i> .
Secondary Action Button	Used for less critical actions or alternative options.
Icon Button	Used for quick actions triggered by icons (e.g., settings, share).

Custom Components

Component	Description
Risk Indicator	A visual cue representing the security risk level of a scanned URL or QR code (e.g., green = safe, yellow = suspicious, red = dangerous).
History Item	Displays previously scanned URLs/QR codes with a summarized security result and timestamp for quick review.
Notification Banner	Shows real-time alerts or updates to inform users about scan results or important security warnings.

UI Screenshots

QR Guard UI Screenshot - Home Screen

QR GUARD

Sign in to your account

USERNAME OR EMAIL

Enter username or email

PASSWORD

Enter your password

Sign In

New to QR Guard?

Create an Account →

Passwords are hashed and stored securely

CREATE ACCOUNT

Join QR Guard to stay protected

USERNAME

ahmed

EMAIL ADDRESS

abdeen@gmail.com

PASSWORD

ahmed12

Medium

CONFIRM PASSWORD

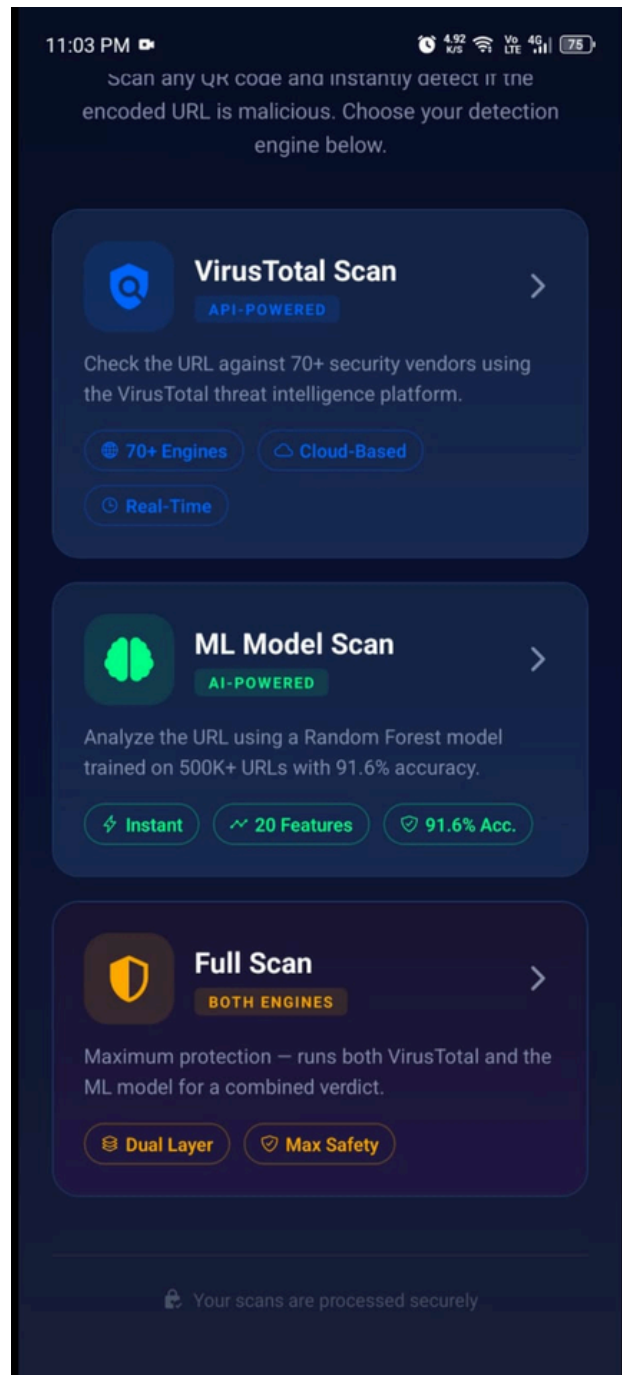
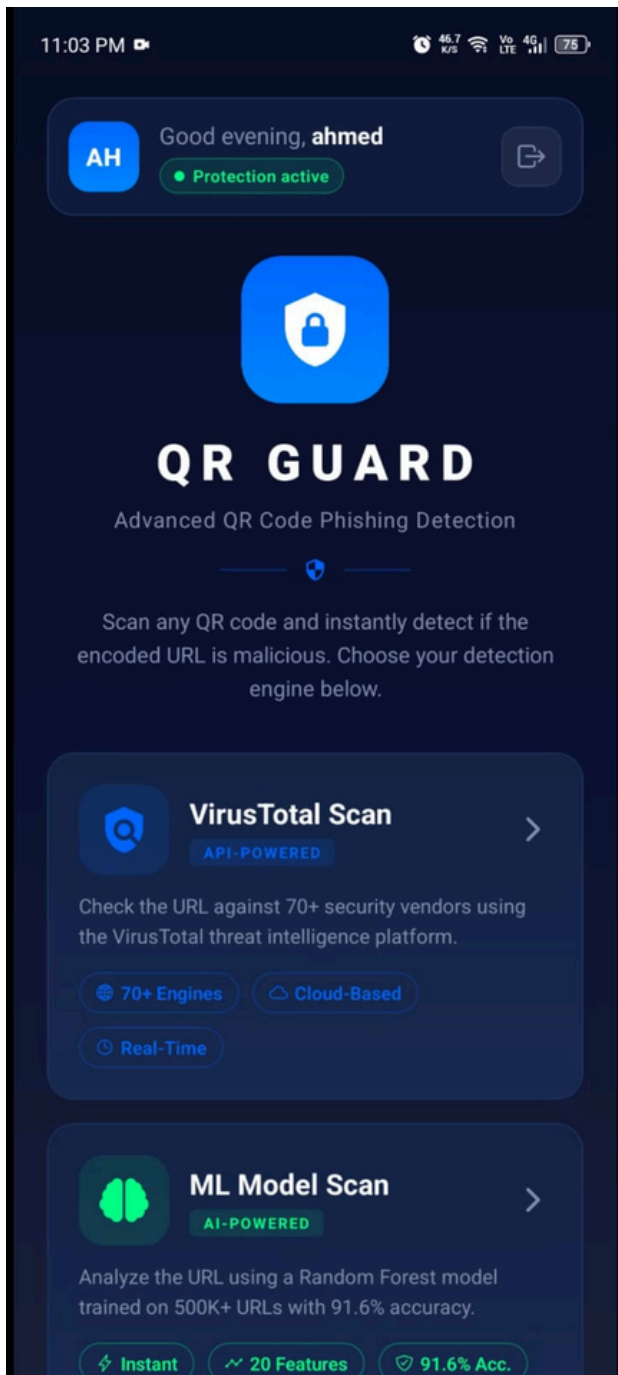
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Create Account

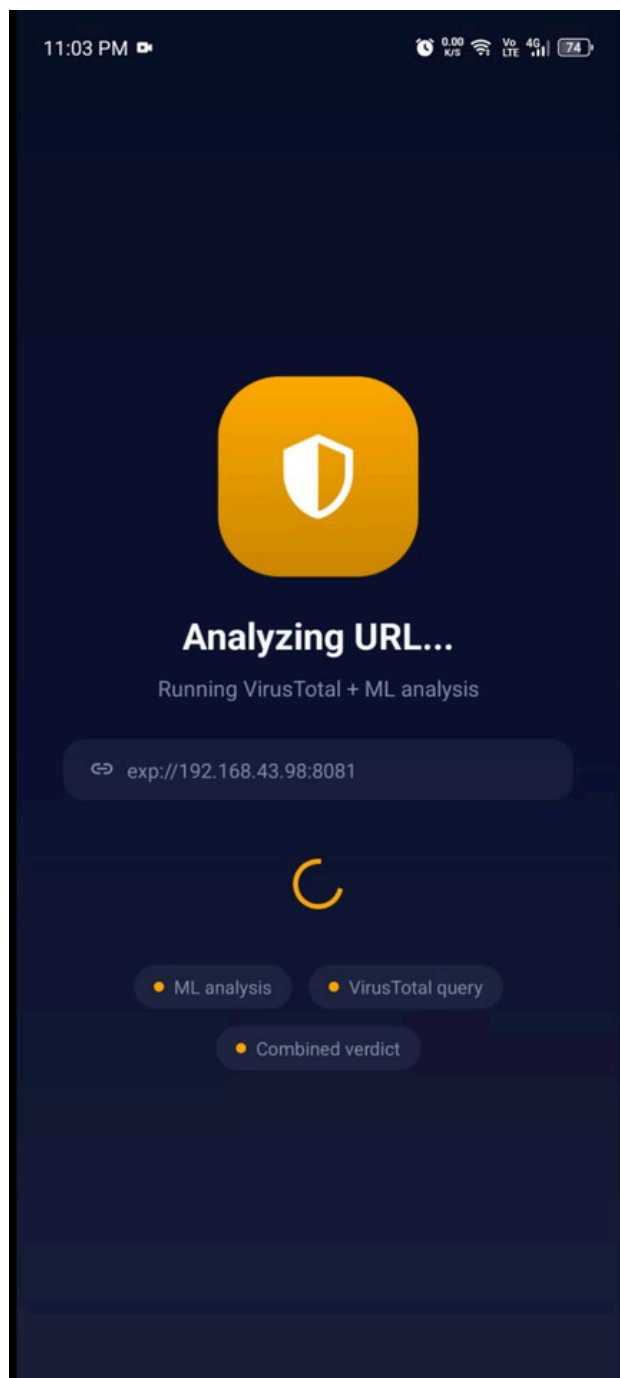
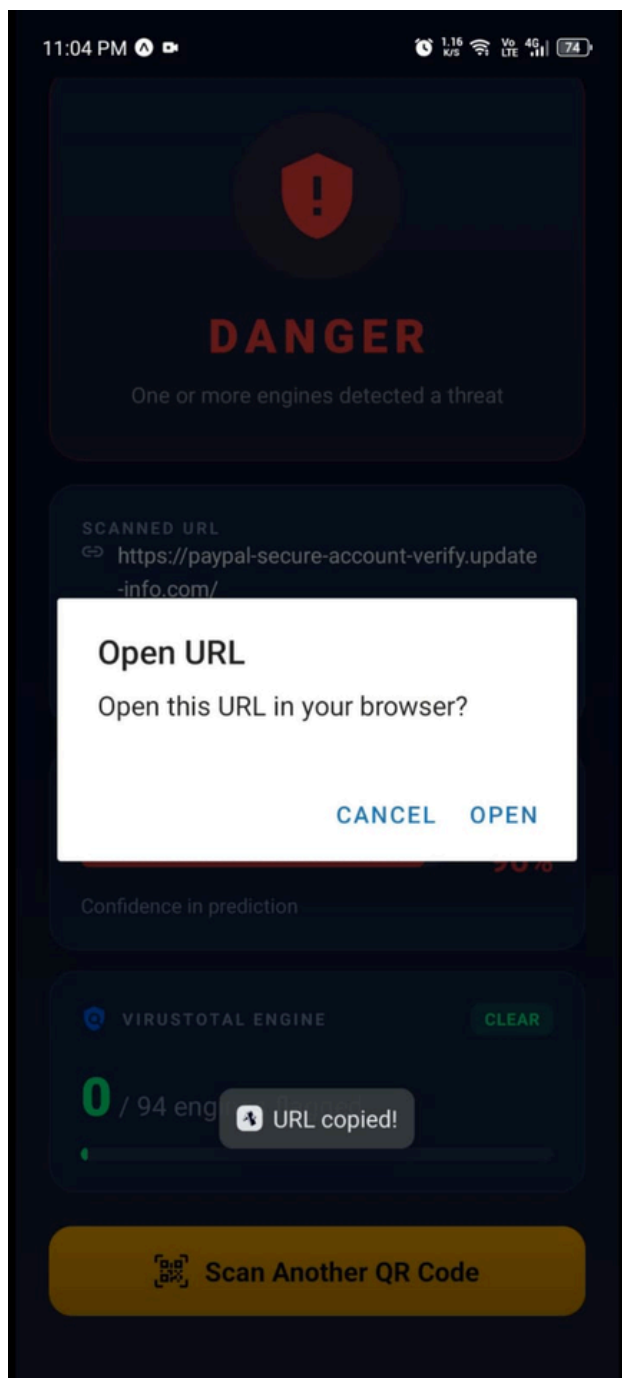
Already registered?

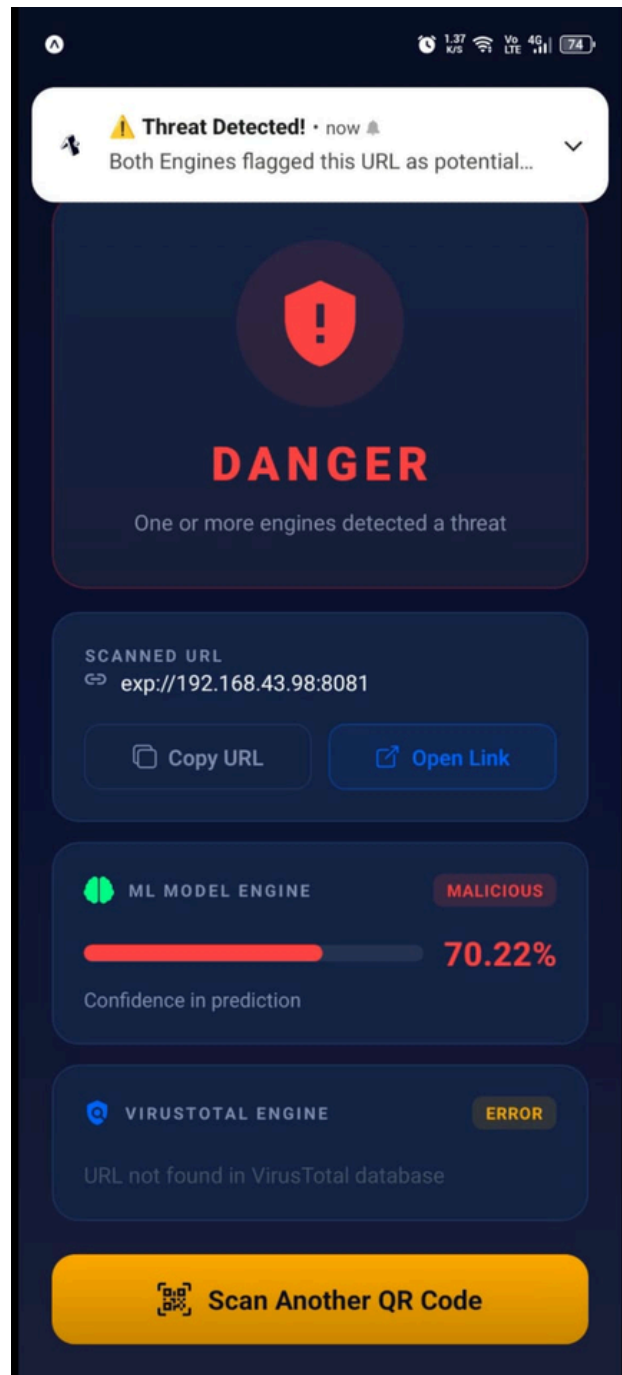
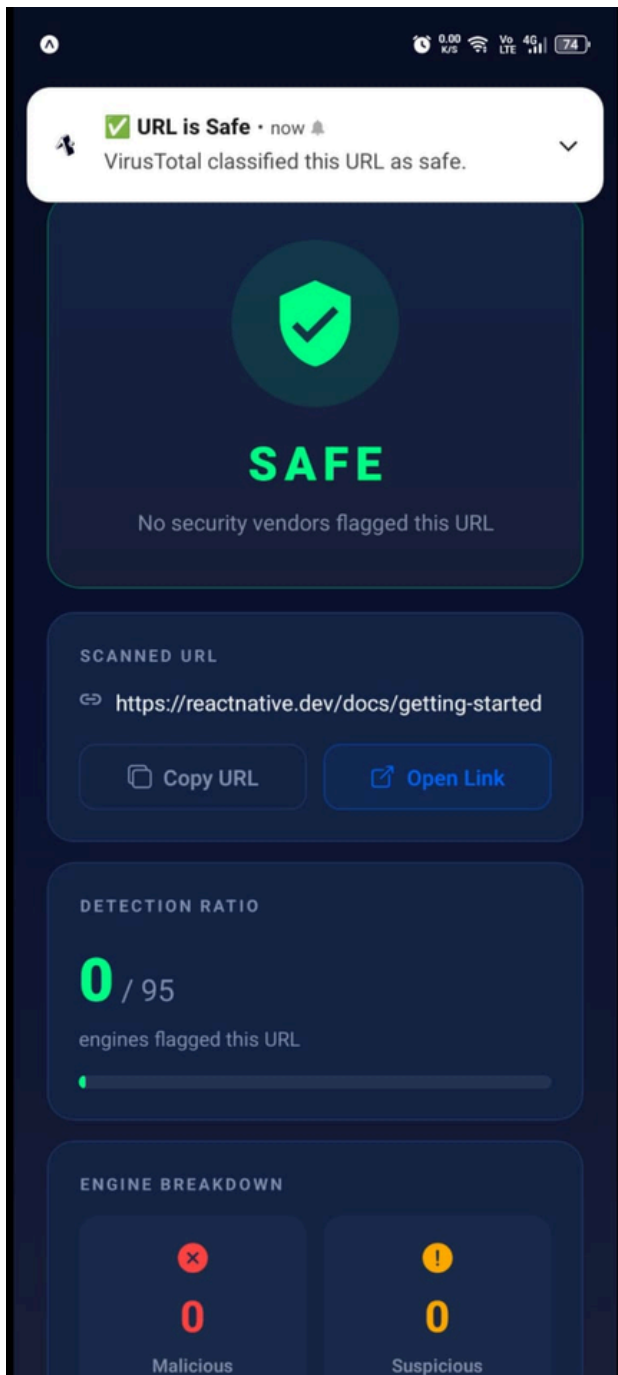
Sign In Instead →

Passwords are hashed with PBKDF2-SHA256

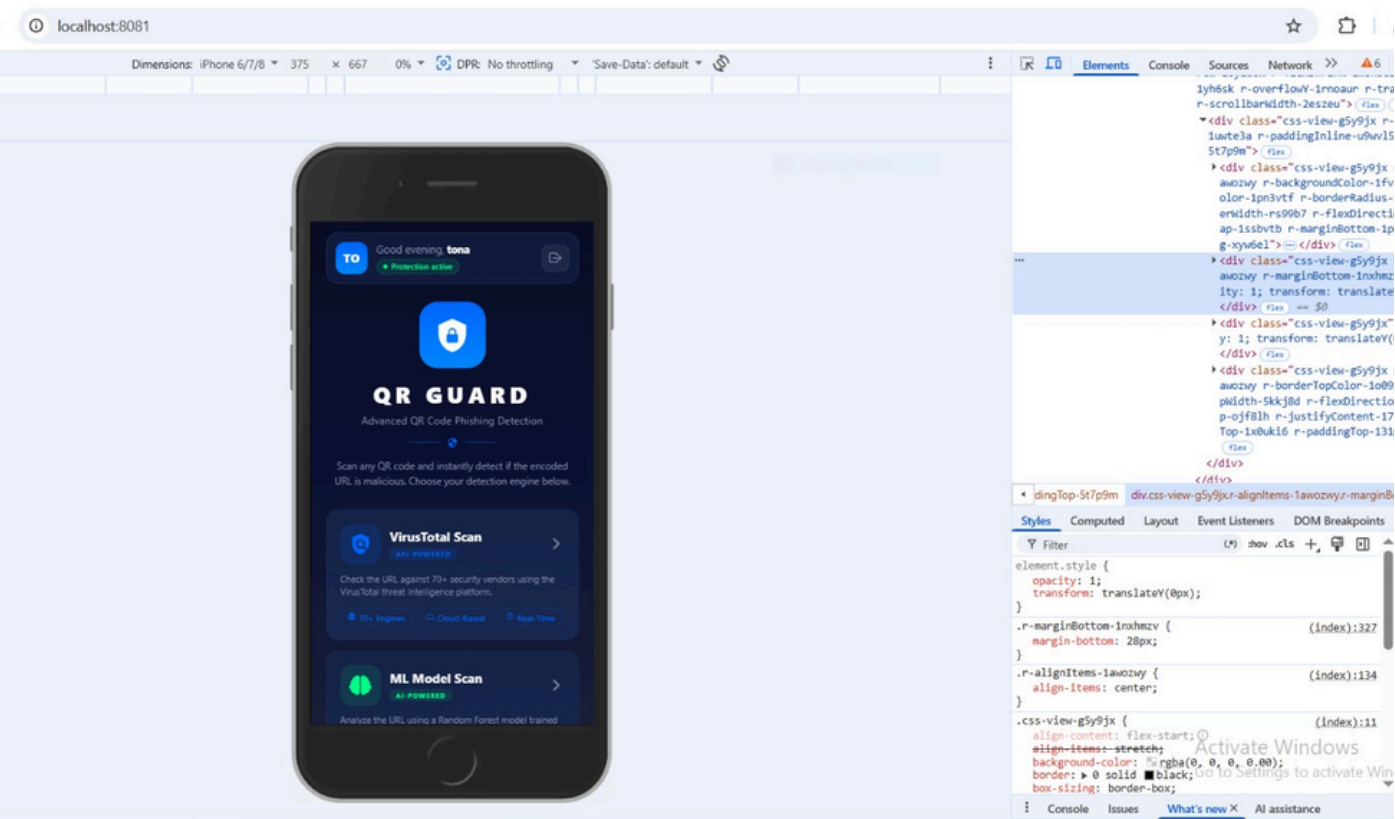


QR Guard UI Screenshot - Scan Result

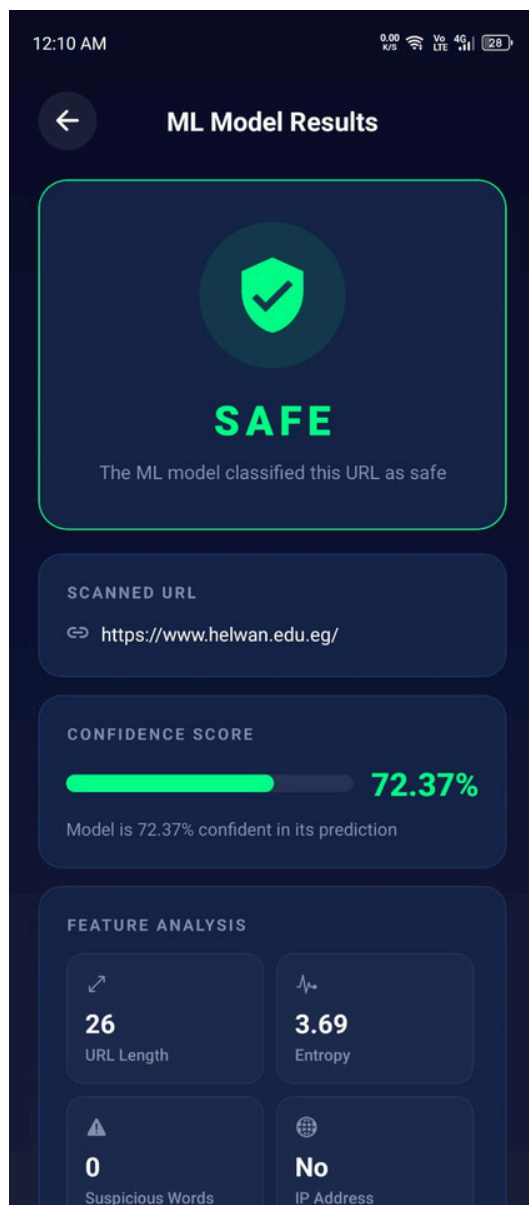




Responsiveness Examples



QR Guard on a laptop Device.



QR Guard on a Mobile Device.

Frontend Technologies

Technology	Purpose
React Native	Cross-platform mobile application development
JavaScript / TypeScript	Programming language
Expo	Development toolkit for React Native
Axios	HTTP client for API requests
Styled-Components	CSS-in-JS library for styling components

Section 03: Backend

Backend Architecture Diagram

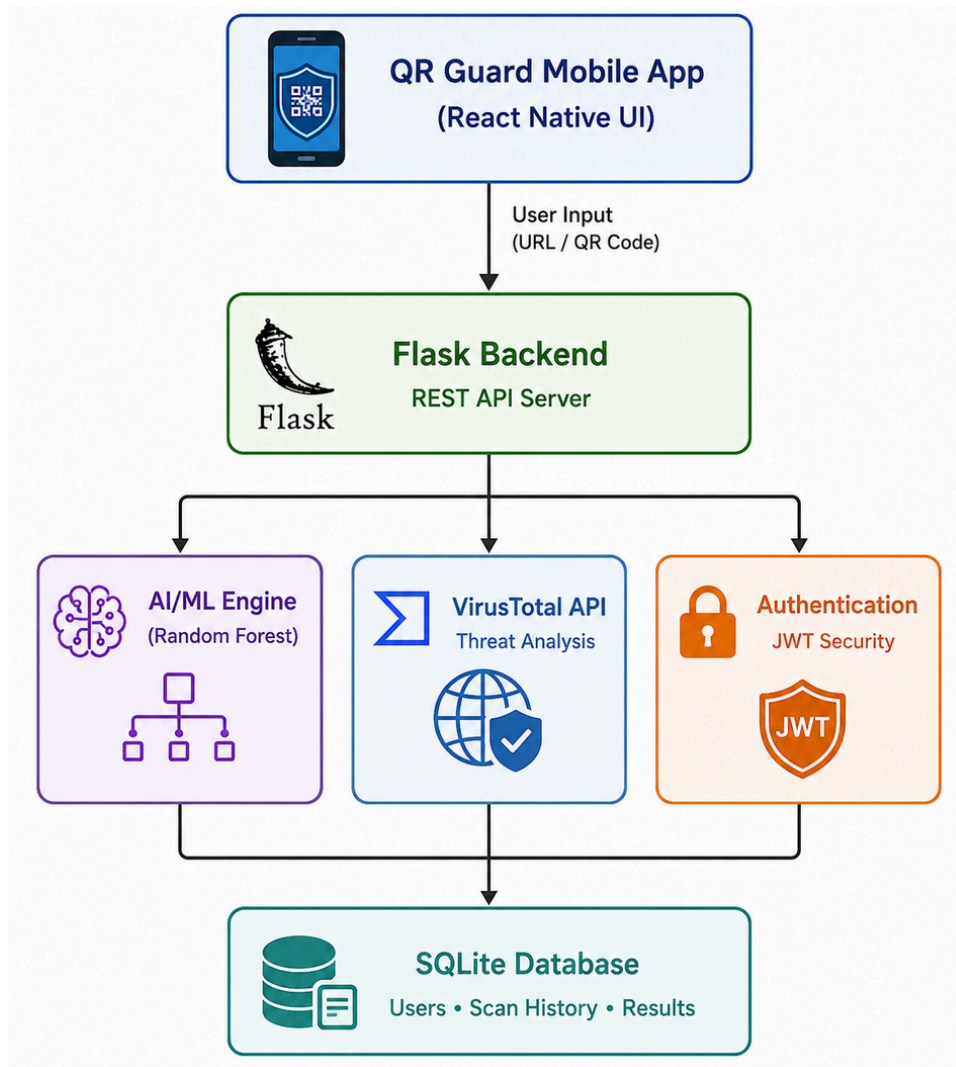


Figure : QR Guard Backend Architecture.

Database Records Table Placeholder

Scan ID	User ID	Scanned URL	Detection Method	Result	Confidence Score	Scan Date
001	101	example.com	AI Model	Safe	92%	08/05/2026
002	101	suspicious-link.net	AI + VirusTotal	Dangerous	87%	08/05/2026
003	102	secure-site.org	VirusTotal	Safe	95%	07/05/2026
004	103	fake-login-page.com	AI Model	Dangerous	90%	07/05/2026

Sample of Stored User Scan History

API Endpoints Table

Endpoint	Method	Description
/auth/register	POST	Register a new user account
/auth/login	POST	Authenticate user and generate JWT token
/auth/me	GET	Retrieve authenticated user profile information
/scan/ml	POST	Submit a URL for machine learning analysis
/scan/virustotal	POST	Submit a URL for VirusTotal reputation analysis
/scan/both	POST	Submit a URL for combined AI and VirusTotal analysis
/scan/history	GET	Retrieve the authenticated user's scan history

Dynamic Content Explanation

The backend dynamically generates and delivers security risk scores and detailed threat analyses based on the input URL or QR code data. This involves real-time interaction with the AI model and the VirusTotal API, processing their responses, and formatting them into a user-friendly output for the mobile application. User-specific data, such as scan history, is also dynamically retrieved and presented.

Backend Technologies

Technology	Purpose
React Native	Cross-platform mobile application development for building the QR Guard user interface
Expo Router	Navigation and screen management within the mobile application
Flask	Backend API development and server-side logic implementation
SQLite	Database management for storing user accounts and scan history
JSON Web Token	Secure user authentication and session management
Random Forest	Machine learning model for URL classification and phishing detection
VirusTotal API	Threat intelligence integration for checking URL reputation and malicious activity
Git	Version control and project collaboration
GitHub	Source code hosting, project management, and documentation
Python Libraries (Pandas, Scikit-learn)	Data preprocessing, model training, and feature extraction

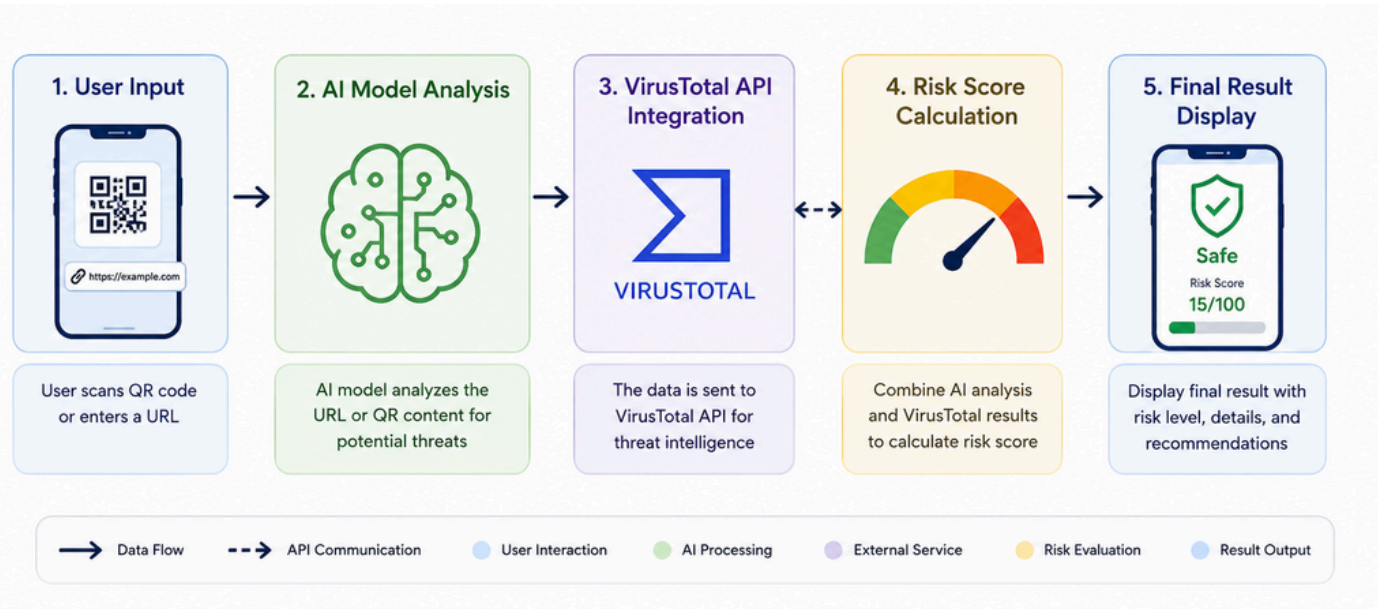
Features

- **Authentication:** Secure user registration and login to manage scan history and personalize experience.
- **Navigation:** Intuitive and easy-to-use interface for seamless movement between scanning, history, and settings.
- **Notifications:** Real-time alerts for scan results, indicating potential threats or safe status.
- **QR Scanning:** Utilizes the device camera to quickly scan and decode QR codes.
- **URL Scanning:** Allows users to input or paste URLs for immediate security analysis.
- **AI Detection:** Employs advanced machine learning models to identify sophisticated and zero-day threats, including phishing attempts and malicious scripts.
- **VirusTotal Integration:** Leverages the VirusTotal API to cross-reference URLs and QR code destinations against a vast database of known malicious sites and files.

- **History:** Stores a log of all scanned URLs and QR codes, providing users with a record of their digital interactions and their security status.

System Flow Diagram

System Flow Diagram Placeholder



Flow Description:

- 1. User Input:** The user scans a QR code or enters a URL.
- 2. Data Processing:** The input is sent to the backend.
- 3. AI Model Analysis:** The AI model analyzes the URL/QR content for malicious patterns.
- 4. VirusTotal API:** The URL is simultaneously checked against the VirusTotal database.
- 5. Risk Score Calculation:** Results from AI and VirusTotal are aggregated to compute a comprehensive risk score.
- 6. Result Display:** The user receives a clear indication of the risk level and any detected threats.

Team Contribution

Member	Role	Contribution
Hassan Adel Hassan	Project Lead & Backend Developer	Overall project planning, backend architecture design, authentication system implementation, API integration.
Eman Mohamed Hamed	UX/UI Designer	User research, survey analysis, empathy map creation, persona development, wireframing, and UI design.
Basma Mohamed Nagib	Frontend Developer	Mobile application development using React Native, UI implementation, navigation setup, responsive design implementation.
Ahmed Ehab mahmoud	Backend & Database Developer	Database management using SQLite, API development, scan history management, backend optimization.
Mariam Maher Mohamed	Machine Learning Engineer	Machine learning model training, URL feature extraction, AI integration, model testing, and threat detection analysis.

Conclusion

QR Guard successfully addresses the growing need for robust mobile cybersecurity by integrating advanced AI capabilities with real-time threat intelligence. The application provides users with an intuitive and powerful tool to navigate the digital world safely, mitigating risks associated with QR codes and URLs. The project demonstrates a comprehensive approach from user experience research to sophisticated backend implementation, delivering a valuable solution for everyday digital security concerns. The chosen technologies and design principles ensure scalability, reliability, and a user-centric experience, positioning QR Guard as a significant asset in personal cybersecurity.

References

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[OWASP Official Website](#)